

RESEARCH ARTICLE

Modeling and Solving Multi-Objective Path Planning Problem for Cooperative Cable-Suspended Load Transportation Considering the Time Variable Risk

AMIR ARSLAN HAGHRAH^{ID}, SEHRANEH GHAEMI^{ID},
AND MOHAMMAD ALI BADAMCHIZADEH^{ID}

Department of Electrical and Computer Engineering, University of Tabriz, Tabriz 5166616471, Iran

Corresponding author: Sehraneh Ghaemi (ghaemi@tabrizu.ac.ir)

ABSTRACT Cooperative load transportation has the potential to revolutionize the urban logistics industry and can also be utilized in rescue operations following natural disasters. Utilizing cooperative cable-suspended load transportation in such applications involves addressing three problems: static path planning, dynamic path planning, and control. In this article, we aim to tackle the first problem, i.e., static path planning, which is largely separate from the other two tasks. In this regard, we propose a model for the offline path planning of the cooperative cable-suspended load transportation system, including practical constraints. Using the proposed model, we can limit parameters such as the speed and acceleration of UAVs, flight altitude, task completion time, and safety distances between UAVs and obstacles, while minimizing the path length and risk. In addition, we propose an improved hybrid fuzzy particle swarm optimization (PSO) and whale optimization algorithm (WOA) to solve the introduced path planning problem efficiently. Furthermore, a chaotic mutation operator prevents premature convergence in the PSO algorithm. To demonstrate the efficiency of the proposed model and algorithm, three case studies inspired by a fire incident in a forest park have been conducted, and the results have been compared with those of seven other well-known algorithms.

INDEX TERMS Cooperative robotics, path planning, multi-objective optimization, heuristic algorithms.

I. INTRODUCTION

Nowadays, UAVs are present in a diverse and wide range of applications, including intelligent farming [1], fast delivery, natural disaster rescue operations, surveillance, reconnaissance and mapping, and communications [2]. One of the principal applications of UAVs is delivery and transportation [3]. Today, many companies in developed countries have started offering fast aerial delivery services using UAVs. Transportation using UAVs is not limited to local courier services but also can be used for emergency uses; for example, delivering medicine to difficult areas or delivering equipment and essentials for survival in disasters [4]. Of course, this innovative transportation method, along with

increasing the speed of service and reducing costs, has some restrictions. These restrictions often include a strict limit on load weight and lower transportation reliability. The cooperative transportation concept has been proposed to address these drawbacks [5]. The practical applications of cooperative load transportation using UAVs are vast and significant, particularly in scenarios requiring the delivery of vital supplies to challenging or dangerous locations. For instance, in the context of forest fire incidents, the ability to transport essential elements such as medical supplies, water, or equipment to firefighters can be critical in supporting emergency response efforts. Additionally, this technology has potential applications in disaster relief, where UAVs can be deployed to deliver aid to areas that are otherwise inaccessible due to damaged infrastructure. Other practical uses include precision agriculture, where UAVs

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can cooperatively transport and distribute resources such as seeds or pesticides, and industrial logistics, where UAVs can be used for tasks like assembling large structures or transporting parts in hard-to-reach areas. The emphasis on path planning within this research is crucial, as it ensures that these UAVs can operate efficiently and safely, navigating complex environments while minimizing risk and optimizing resource usage. This research, therefore, holds considerable potential for enhancing the effectiveness and safety of UAV operations in a variety of critical applications.

Based on the previous research, UAV path planning problems can be categorized as path planning for a single UAV, multiple UAVs with separate tasks, and multiple cooperating UAVs. The problems in the first and second categories are mostly used in reconnaissance applications. In the second category, in addition to the path planning of each UAV, an optimal task assignment procedure should also be done, which further complicates the problem [6]. The third category that has recently come to the fore is UAV path planning for cooperative tasks. As one of these tasks, path planning of UAVs in a cable-suspended load transportation system has rarely been studied. The following is a review of the studies concerning the path planning problem for UAVs.

A recent review of the reinforcement learning-based air combat maneuver decision-making has been done by Wang et al. [7], which is useful for this topic. Numerous studies have addressed solving the path planning problem for UAVs [8]. Chang et al. have used dynamic windows approach along with Q-learning for robot path planning and demonstrated efficiency of their proposed method encountering complex unknown environments [9]. Pei et al. have studied mobile robot path planning in dynamic environments using Dyna-Q algorithm [10]. The proposed algorithm integrates heuristic search strategies, a simulated annealing mechanism, and reactive navigation principles into the Q-learning framework within the Dyna architecture. The primitive methods used for solving UAVs' path planning, similar to many other path planning problems, are the A-star algorithm [11], Dijkstra's algorithm [12], Q-learning [13], and dynamic programming [14]. Wang et al. have studied the path planning problem for urban public transportation using dynamic programming [15]. Li et al. have introduced a new adaptive dynamic programming for mobile robots path planning [16]. They have designed a new performance index function for minimizing the path length and avoid collision with obstacles. Mathematical programming in motion planning problems have been studied by Abichandani et al. [17]. They have addressed the multi-vehicle motion planning problem considering various challenging constraints. Mixed integer programming have also been utilized extensively for solving path planning problems [18].

These classical optimization methods are not well-suited for nonlinear problems with complex, non-convex constraints, which often result in NP-hard optimization challenges [19], [20]—issues that are particularly common

in UAV path planning. That is why intelligent algorithms have become a reliable tool utilized in many studies in this field [8]. Genetic Algorithm (GA) is a basis for numerous and various methods implemented for the path planning of robots [21]. In [22], the authors studied a simplified 2D UAV path planning problem and solved it using an improved genetic algorithm. They have used an adjunct population that helps to maintain the dispersion of individuals. In [23], the authors have proposed an enhanced genetic algorithm for multi-objective multi-robot path planning problems, which can be extended to the path planning of UAVs. Also, in [24], the authors have proposed a modified genetic algorithm for Bezier curves-based path planning of UAVs.

In [25], the authors have used a combination of GA and ACO algorithms for UAV path planning in infrastructure monitoring applications. They have provided an optimal path planning mechanism based on environmental information such as prohibited zones in space, risk, sensor deployment statistics, and geo-locational conditions. In [26], Ma et al. have addressed the concurrent path planning and duty assignment problem of UAVs as a Multiple-TSP problem. Also, the paper focuses on finding the minimum required number of UAVs to finish all tasks in a limited time. A combination of the GA and cluster algorithm has been proposed to solve the problem. A gradient-based sequence minimal optimization method has been proposed in [27] for multi-UAV trajectory planning. The authors have utilized a time segmentation-based method for simplifying the cost function calculation. The other algorithm, which also is used frequently in path planning problems, is Ant Colony Optimization (ACO) algorithm [28], [29], [30]. Another study focusing on the less investigated areas of path planning of UAVs is [31]. In this reference, the authors have tried to solve issues like slow convergence speed, low searching efficiency of previous algorithms, and collisions between UAVs and the obstacles. The proposed improved ACO algorithm, called MACO, has been used for generating collision-free trajectories. Finally, after enhancing the trajectories, they have been smoothed using the inscribed circle (IC) method.

In [32], Al Hilli et al. have studied the general path planning problem in stochastic quasi-dynamic environments using the PSO algorithm. Sonny et al. have also proposed a modified PSO algorithm for UAV path planning [33]. Hybrid algorithms, with their special capability of handling complex optimization tasks [34], are also widely used for solving path planning problems. In [35], Abhishek et al. proposed two hybrid PSO-HSA and PSO-GA algorithms for the 3D path planning of UAVs. In addition to these two methods, they have made a comprehensive comparison between different optimization methods for solving the UAV path planning. The results have shown that hybrid algorithms can improve the results of path planning problems. Many other nature-inspired algorithms have been used for solving path planning, which the Firefly Algorithm (FFA) [36], Whale Optimization Algorithm [37], Multiverse Optimizer Algorithm (MVO) [38], Bat Algorithm (BA) [39], Gray Wolf Optimization

(GWO) algorithm [40], and Cuckoo Search Optimization (CSO) algorithm [41] are among them.

In this paper, we formulate the offline path planning problem for cooperative load transportation. The proposed formulation is capable of considering practical constraints for having control over many parameters. The ability to handle the maneuvering time, the tension of the cables, speed limits and acceleration limits for load and UAVs, risk of crossing different areas, flight altitude, avoidance of various obstacles, and preventing inter-UAV collision, along with minimizing the path length are characteristics of the proposed model. We consider the path planning problem for cooperative load transportation a multi-objective optimization problem with the path length and risk as objectives. This optimization problem is solved using eight algorithms, Particle Swarm Optimization (PSO) [42], Genetic Algorithm (GA) [43], Firefly Algorithm (FFA) [44], Gray Wolf Optimization (GWO) [45], Whale Optimization Algorithm (WOA) [46], Human Evolution Optimization (HEO) [47], Improved PSO [48], and the proposed Hybrid Fuzzy WOA-PSO. Recent years, there were efforts for using PSO and WOA algorithms in a hybrid scheme [49], [50]. Najm et al. have utilized hybrid WOA-PSO algorithm for path planning of wheeled mobile robots. They have compare the achieved results with conventional GWO, WOA, and PSO algorithms [51]. Also, Trivedi et al. have proposed a hybrid WOA-PSO algorithm for optimization of global numerical functions. In their proposed method, PSO algorithm is responsible for exploitation phase, and the WOA carries out the exploration phase [52]. Nasrollahzadeh et al. have used a hybrid WOA-PSO algorithm for optimal motion sensor placement which is crucial in smart homes and intelligent environments [53]. The main reason for such hybridization can be found in the nature of these two algorithms. The main drawbacks concerning the PSO algorithm include its tendency to cover a limited search space and its struggles with high-dimensional problems or complex objective functions, primarily because it relies on a constant inertia weight. In contrast, WOA is capable of exploring a broader and more uncertain space by employing helical logarithmic functions. By combining these two algorithms, a hybrid approach can be developed that mitigates their individual weaknesses while enhancing their strengths [53]. In our hybridization scheme, in addition to utilizing a fuzzy logic-based conscious method for choosing the algorithm to be applied, we have improved the PSO algorithm by entering the best experienced velocities of the swarm and particles into the position update procedure. Also, we have utilized a chaotic random walk mutation operator to the PSO algorithm to avoid premature convergence. Furthermore regarding the WOA algorithm, we set a parameter based on the iteration index, giving it adaptation to solutions enhancement. Considering all these efforts, we have proposed a unique improved hybrid fuzzy PSO-WOA algorithm, which can successfully utilized for solving complex non-linear and non-convex problems.

The main contributions of this paper can be summarized as:

- Development of a comprehensive mathematical model for multi-objective path planning in cooperative UAV-based cable-suspended load transportation incorporating path risk into the optimization process.
- Introduction of a novel hybrid algorithm combining a fuzzy logic-based decision system with Whale Optimization Algorithm (WOA) and Particle Swarm Optimization (PSO), including an innovative chaotic walk-based mutation operator and the use of stored best velocities to enhance convergence.

The proposed method demonstrates superior performance compared to other state-of-the-art algorithms.

We have designed two case study for simulating the duty of carrying vital items to firefighters in a forest. In the first case there is only one objective, minimizing the path. But in the second case the risk should also be minimized along with minimizing the path length. Note that in fire incidents, the wind can transfer the center of the fire gradually. So, in this case, the time to cross different areas will also be critical. This case study considers the risk source moving as a function of time. The simulation results approve the efficiency of the proposed model and show that the proposed hybrid optimization algorithm can solve the defined path planning problems better than the other seven algorithms.

The rest of the paper is organized as follows: Section II introduces the proposed problem model. The proposed hybrid fuzzy WOA-PSO algorithm and constraints handling method used in implementations are described in section III. Section IV represents the simulations and compares the results achieved by different optimization algorithms. Finally, the paper's conclusions are presented in section V.

II. PROBLEM FORMULATION

Figure 1 illustrates the overall path planning and control problem for the cooperative cable-suspended load transportation system. As can be seen, the first step is offline path planning, which is done considering the system constraints and some objective functions. After this step, the designed path enters into the online path planning block. Online path planning prevents collisions at a systemic level and between system and dynamic obstacles. The inputs of the online path planning block are the designed path in the offline path planning process and the position of the load and each quadrotor. It is assumed that the online path planning block has enough information about the obstacles in the workspace. Unlike the offline path planning block, the online path planning block depends entirely on system dynamics. The last block is the tracking control, which determines the thrust of each quadrotor based on the output path achieved from the second block. Also, in the third block, the thrust vector determines the desired attitude of quadrotors, which must be tracked. As has been said before, this paper aims to model the offline path planning problem for a cooperative cable-suspended load transportation system and solve it for different scenarios.

Cooperative Load Transportation System

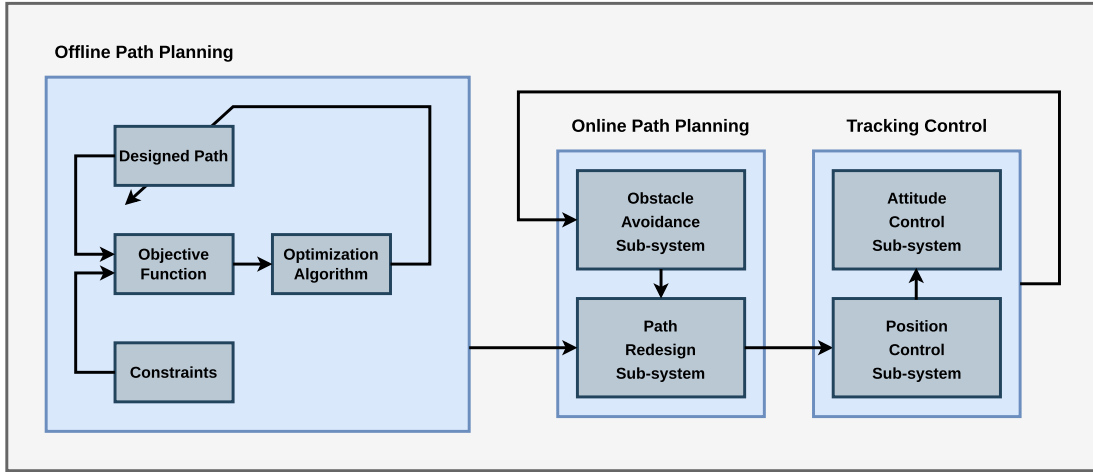


FIGURE 1. The overall path planning and control problem for the cooperative cable-suspended load transportation system.

Therefore, it will not directly deal with the dynamics of the system. However, the speed and acceleration constraints that arise from the system dynamics will be among the inputs of the path planning problem.

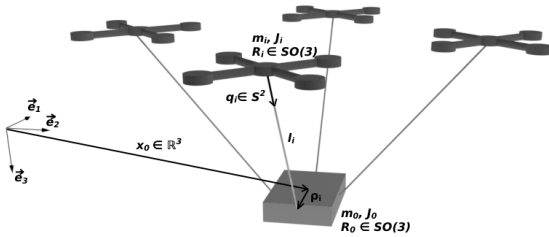


FIGURE 2. An overview of the cooperative cable-suspended load transportation system.

In the cooperative cable-suspended load transportation path planning problem, we are looking to find the shortest possible path for the load, starting from the origin and ending at the destination. The designed path must be collision-free and also meet some other expectations. These expectations can be expressed in terms of time, risk, etc. The point that makes the cooperative load transportation path planning problem different and more complicated compared with conventional robot path planning problems is the physical interconnection of the sub-systems. Considering that there is no comprehensive model for this problem, we provide a mathematical model for cooperative cable-suspended load transportation path planning in this paper.

An overview of the cooperation transportation of the cable-suspended load is illustrated in Figure 2. As can be seen, the position of the load with respect to the fixed coordinate system is named $\underline{x}_0$, the number of cooperating UAVs is N_q , and positions of the UAVs are defined as $\underline{x}_i$, $i = 1, \dots, N_q$. The length of the cable connecting the load to the i^{th} UAV is

indicated by l_i . $q_i \in S^2$ indicated the direction of the links from the i^{th} UAV toward the connection point of the link and the load, and S^2 is defined as the set $\{q \in \mathbb{R} \mid \|q\| = 1\}$. Moreover, ρ_i is the vector connecting the center of mass of the load to the connection point of the i^{th} cable to the load. The m_0 and J_0 indicate the mass and inertia matrix of the load. Also, m_i and J_i are the mass and inertia matrix of the i^{th} UAV. $R_0 \in SO(3)$ indicates the rotation matrix of the load, and $R_i \in SO(3)$ indicates the rotation matrix of the i^{th} UAV. $SO(3)$ is the special orthogonal group and is defined as the set $\{R \in \mathbb{R}^{3 \times 3} \mid R^T R = I, \det[R] = 1\}$.

In this article, the method for avoiding collisions is quantifying the environment using penalty values. So the obstacles and coordinates near them would have large penalty values in contrast with the other free zones. With these assumptions, the path planning problem objective for achieving the shortest path would be as below:

$$f_1(\underline{x}_0(t)) = \int_{t_0}^{t_e} \|\dot{\underline{x}}_0(t)\| dt \quad (1)$$

In which, $\underline{x}_0(t)$ is the position of the load. Also, t_0 and t_e indicate the start and end times of the maneuver. Notice that, without affecting the generality, t_0 is assumed 0. There are also physical constraints that must be met.

The first constraint is starting position of the load and its destination, which can be formulated as below:

$$\underline{x}_0(t_0) = \underline{x}_0^* \quad (2)$$

$$\underline{x}_0(t_e) = \underline{x}_e^* \quad (3)$$

The second constraint is the upper limit of the load speed and acceleration in the environment:

$$\|\dot{\underline{x}}_0(t)\| \leq v_0^{\max} \quad (4)$$

$$\|\ddot{\underline{x}}_0(t)\| \leq a_0^{\max} \quad (5)$$

The third constraint of the system is avoiding inter-UAV collisions, which can be written over the distance of the UAVs:

$$\| \underline{x}_i(t) - \underline{x}_j(t) \| \geq d_i + d_j + \text{safety distance},$$

$$i, j = 1, \dots, N_q, i \neq j \quad (6)$$

The safe zone around the obstacle is based on an assumption intended to reduce the impact of unmodeled details and uncertainties in the environment. This assumption allows for a more practical and robust solution by providing a buffer that accounts for possible variations in obstacle shape or movement. The specific size of the safe zone was chosen to ensure sufficient safety margin without overcomplicating the model. Also, the UAVs must avoid collision with obstacles. This issue can be modeled as below:

$$\forall t, \underline{x}_0(t), \underline{x}_i(t) \notin \text{Obstacles} + \text{safety distance}, i = 1, \dots, N_q \quad (7)$$

Another important constraint is the minimum and maximum altitude of the system, which must lay in a specific interval. It can be achieved by limiting the load altitude as the formula below:

$$z_0^{\min} \leq z_0(t) \leq z_0^{\max} \quad (8)$$

Notice that based on the definitions, the position of the quadrotors based on other parameters of the system can be calculated as below:

$$\underline{x}_i(t) = \underline{x}_0(t) + R_0 \rho_i - l_i q_i(\theta_i(t), \phi_i(t)), i = 1, \dots, N_q \quad (9)$$

As can be seen, the q_i is written as a function of $\theta_i(t)$ and $\phi_i(t)$. The $\theta_i(t)$ is assumed as the angle between the image of the q_i over the $x - y$ plane and the x -axis. Also, the $\phi_i(t)$ is assumed as the angle between the q_i and $x - y$ plane. By these definitions, the angles would be limited as below:

$$0 \leq \theta_i(t) < 2\pi, 0 \leq \phi_i(t) \leq \frac{\pi}{2}, i = 1, \dots, N_q \quad (10)$$

Also, the explicit relation between the q_i , and these two angles would be:

$$q_i(\theta_i(t), \phi_i(t)) = \begin{bmatrix} \cos(\phi_i(t)) \cos(\theta_i(t)) \\ \cos(\phi_i(t)) \sin(\theta_i(t)) \\ \sin(\phi_i(t)) \end{bmatrix},$$

$$i = 1, \dots, N_q. \quad (11)$$

The tension of the cables is a function of the $\phi_i(t)$. To achieve minimum tension in the cables, we should minimize the amplitude of the image of the cable tension vector on the x - y plane. Ideally, this would result in each quadrotor applying only a vertical force to counteract gravity. So, we are looking to bring the ϕ_i angle closer to $\frac{\pi}{2}$, while avoiding collisions and maintaining the safety distance between UAVs.

$$\phi_i(t) \rightarrow \frac{\pi}{2}, i = 1, \dots, N_q, t_0 \leq t \leq t_e \quad (12)$$

This issue can be entered into the model by a penalty as the integral of the $\frac{\pi}{2} - \phi_i(t)$. While adding this penalty to the

objective function of the optimization problem, we should consider that the associated penalty must be less than collision avoidance penalties. So, we assume small values as the coefficients of this integral.

$$\int_{t_0}^{t_e} \left[\frac{\pi}{2} - \phi_i(t) \right] dt \quad (13)$$

For making the mechanical constraint over the link length, UAV position, and load position satisfied, the position of the UAVs can be calculated using equation (9). So, the speed and acceleration of the UAVs can be limited by limiting the first and second derivation of the angles $\theta_i(t)$ and $\phi_i(t)$:

$$|\dot{\theta}_i(t)| < \dot{\theta}_i^{\max}, |\dot{\phi}_i(t)| \leq \dot{\phi}_i^{\max}, i = 1, \dots, N_q \quad (14)$$

$$|\ddot{\theta}_i(t)| < \ddot{\theta}_i^{\max}, |\ddot{\phi}_i(t)| \leq \ddot{\phi}_i^{\max}, i = 1, \dots, N_q \quad (15)$$

Finally, the last constraint to introduce is the mission time limits. In many applications, we need to finish the assigned task during a specific time interval, which can be entered into the problem model as below:

$$t^{\min} \leq t_e - t_0 \leq t^{\max} \quad (16)$$

Considering these definitions, the problem is completed by assuming the minimization of path length as the objective function. Besides, we can also add minimization of the risk as the second objective of the path planning to the model as below:

$$f_2 = \int_{t_0}^{t_e} \gamma(\underline{x}_0(t), t) dt \quad (17)$$

In which $\gamma(\underline{x}, t)$ indicates the distribution of the risk in the environment. γ is a function of both position and time parameters. So, in the path planning problem, we can consider the risk patterns changing over time, too. Finally, the optimization problem for the cooperative cable-suspended load transportation path planning can be summarized as Table 1. The parameters utilized in Table 1, are all introduced in Table 2 for better and flash undersandation of the problem.

A. MODELING THE OBSTACLES

As mentioned, the environment is quantified using penalty values indicating the presence of obstacles. So, multi-criteria functions assigning a number in inverse relation to the distance from the obstacle can model the penalty distribution. For example, the function modeling a cylindrical obstacle is as below:

$$s_1 = \| [a - x, b - y] \| \quad (18)$$

$$s_2 = |c - z| \quad (19)$$

$$\psi(x, y, z, t) = \begin{cases} k & s_1 \leq r, s_2 \leq \frac{h}{2} \\ ke^{-\frac{(s_2 - \frac{h}{2})^2}{\sigma^2}} & s_1 < r, s_2 > \frac{h}{2} \\ ke^{-\frac{(s_1 - r)^2}{\sigma^2}} & s_1 > r, s_2 < \frac{h}{2} \\ ke^{-\frac{\max(s_2 - \frac{h}{2}, s_1 - r)^2}{\sigma^2}} & s_1 > r, s_2 > \frac{h}{2} \end{cases} \quad (20)$$

TABLE 1. The multi-objective optimization problem for path planning of cooperative cable-suspended load transportation system.

Minimize	$f_1(\underline{x}_0(t)) = \int_{t_0}^{t_e} \ \dot{\underline{x}}_0(t)\ dt$
	$f_2(\underline{x}_0(t)) = \int_{t_0}^{t_e} \gamma(\underline{x}_0(t), t) dt$
Subject to	$\underline{x}_0(t_0) = \underline{x}_0^*, \underline{x}_0(t_e) = \underline{x}_e^*$
	$\ \dot{\underline{x}}_0(t)\ \leq \nu_0^{\max}, \ \ddot{\underline{x}}_0(t)\ \leq \alpha_0^{\max}$
	$\ \underline{x}_i(t) - \underline{x}_j(t)\ \geq d_i + d_j + \text{safety distance}$
	$\forall t, \underline{x}_0(t), \underline{x}_i(t) \notin \text{Obstacles} + \text{safety distance}$
	$z_0^{\min} \leq z_0(t) \leq z_0^{\max}$
	$0 \leq \theta_i(t) < 2\pi, 0 \leq \phi_i(t) \leq \frac{\pi}{2}$
	$\underline{x}_i(t) = \underline{x}_0(t) + R_0 \rho_i - l_i q_i(\theta_i(t), \phi_i(t))$
	$q_i(\theta_i(t), \phi_i(t)) = \begin{bmatrix} \cos(\phi_i(t)) \cos(\theta_i(t)) \\ \cos(\phi_i(t)) \sin(\theta_i(t)) \\ \sin(\phi_i(t)) \end{bmatrix}$
	$\phi_i(t) \rightarrow \frac{\pi}{2}$
	$ \dot{\theta}_i(t) < \dot{\theta}_i^{\max}, \dot{\phi}_i(t) \leq \dot{\phi}_i^{\max}$
	$ \ddot{\theta}_i(t) < \ddot{\theta}_i^{\max}, \ddot{\phi}_i(t) \leq \ddot{\phi}_i^{\max}$
	$i = 1, \dots, N_q,$
	$t^{\min} \leq t_e - t_0 \leq t^{\max}$

In which the coordinates of the point to evaluate the penalty is $[x, y, z]$. $[a, b, c]$ determines the center of mass of the cylindrical obstacle. Also, r and h indicate the radius and height of the obstacle, respectively. For a better insight into written formulas, we have generated a random environment using cylindrical obstacles and plotted the penalty distribution. To better understand this 3D penalty distribution, it can be illustrated by 2D plots, considering the z coordinate as some fixed values, which looks like slicing the 3D space. Figure 3 demonstrates the penalty distribution over the random environment in different slices of z axes. Notice that all the obstacles are cylindrical shapes with $r = 10$, $h = 15$, and $\sigma = 10$. In this figure, the radius of the circles is larger than the radius of the obstacles creating a safe distance for avoiding collisions. The exponential decrement of penalty in the vicinity of obstacles causes such penalty concentration. For $z = 20$, and 25 , the slicing passes upper than the obstacles' top. So there is a drop in the penalty amount.

The penalty value for avoiding obstacles can be calculated using the formula below, in which N_o indicates the number of obstacles, and ψ_i represents the penalty distribution corresponding with i^{th} obstacle:

$$\sum_{i=1}^{N_o} \int_{t_0}^{t_e} \psi_i(x(t), y(t), z(t)) dt \quad (21)$$

In this study, we use cylindrical shape to model the obstacles, but any type of obstacle by creating a proper function for representing it can be used. The choice of a cylindrical shape for obstacles in our study is motivated by its relevance to common real-world scenarios, such as poles, trees, and towers, which are frequently encountered in urban and natural environments. The cylindrical shape

simplifies the mathematical modeling of obstacles while still providing a realistic and challenging environment for path planning. This shape impacts the path planning by requiring the algorithm to navigate around objects with a uniform cross-section, which tests the algorithm's ability to avoid obstacles while optimizing the path.

B. MODELING THE RISK

As mentioned, the risk distribution over the problem space can be defined as a function of coordinates and time. There is no restriction on defining such a function. However, in this paper, the risk sources are defined using a Gaussian function as below:

$$\gamma_{\text{source}}(x, y, z, t) = ke^{-\frac{(x_r(t)-x)^2}{c_x^2(t)} - \frac{(y_r(t)-y)^2}{c_y^2(t)} - \frac{(z_r(t)-z)^2}{c_z^2(t)}} \quad (22)$$

In which the $x_r(t)$, $y_r(t)$, and $z_r(t)$ are coordinates of the center of the risk source. Also, the coefficients $c_x(t)$, $c_y(t)$, and $c_z(t)$ indicates the spread of the risk in the direction of x , y , and z axes. The representation of the risk distribution in our model is based on the summation of introduced 3D time-variable Gaussian functions. Each 3D Gaussian function allows us to assume a risk source that is maximized at the center point and decreases as it spreads out from that point. Both the center and the spread of the risk along different axes can be treated as time-variable functions, providing considerable flexibility. Summing up these functions enables us to model complex risk distribution scenarios. This approach was chosen because it allows us to effectively quantify and incorporate environmental hazards into the path planning process. By modeling risk as a distribution, we can more accurately represent the variability and uncertainty present in real-world scenarios, making the planned paths more robust and reliable. The relevance of this representation lies in its ability to penalize risky paths, guiding the robot to safer routes that minimize potential harm or failure, which is crucial for applications in unpredictable environments.

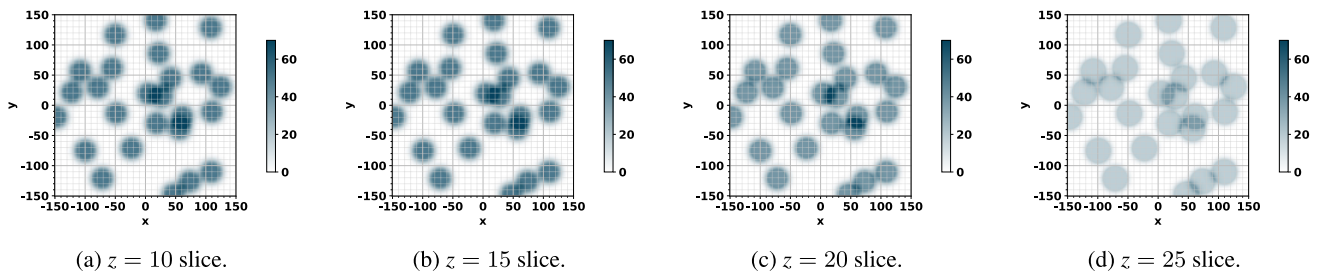
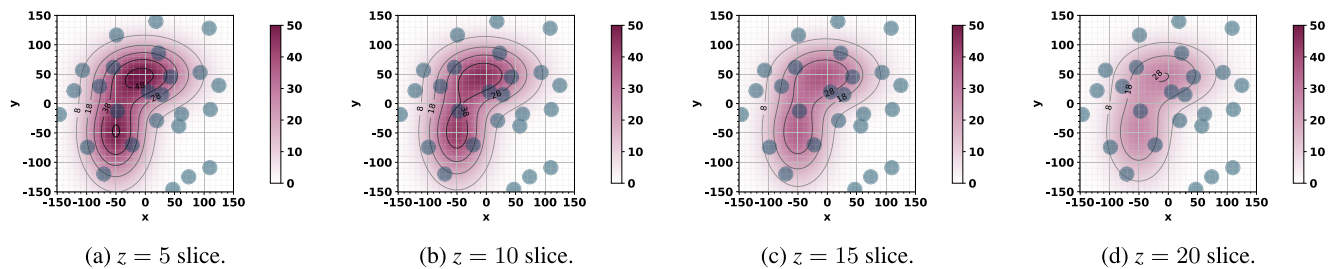
As an example, the risk distribution in an environment with random cylindrical obstacles is defined as the summation of two risk sources as below:

$$\gamma(x, y, z, t) = e^{-\frac{(-50-x)^2}{50^2} - \frac{(-50-y)^2}{75^2} - \frac{(0-z)^2}{25^2}} + e^{-\frac{(0-x)^2}{75^2} - \frac{(50-y)^2}{50^2} - \frac{(0-z)^2}{25^2}} \quad (23)$$

Multiple slices of the risk distribution over an environment with random obstacles, achieved by the formula (23), are represented in Figure 4. The change in the risk distribution over the $x - y$ plane can be seen in the figure. Also, the effect of altitude on the risk distribution is understandable by putting together the four plots presented. Notice that the number obtained by integrating the risk distribution on a given path corresponds to the risk of tracking that path. The higher number, the more likely the mission is to be fallen.

TABLE 2. Definition of the parameters used in the modeling.

Parameter	Unit	Description
$\underline{x}_0(t)$	(m)	The payload position as a function of time.
$\underline{x}_i(t)$	(m)	The i^{th} quadrotor's position as a function of time.
t_0	(s)	The start of the planning time horizon.
t_e	(s)	The end of the planning time horizon.
$\underline{x}_0^*$	(m)	The initial position of the payload at t_0 .
$\underline{x}_e^*$	(m)	The final position of the payload at t_e .
$\nu_0^{\max}$	(m/s)	The maximum limit of the payload velocity.
$\alpha_0^{\max}$	(m/s ²)	The maximum limit of the payload acceleration.
d_i	(m)	The diameter of the i^{th} quadrotor.
Safety distance	(m)	The distance quadrotors should maintain from other objects.
$z_0^{\min}$	(m)	Minimum limit of the payload altitude during the mission.
$z_0^{\max}$	(m)	Maximum limit of the payload altitude during the mission.
$\Theta_i(t)$	(rad)	The polar angle in spherical coordinate system defining the position of the i^{th} quadrotor.
$\dot{\Theta}_i^{\max}$	(rad/s)	The first derivative of the polar angle of the i^{th} quadrotor.
$\ddot{\Theta}_i^{\max}$	(rad/s ²)	The second derivative of the polar angle of the i^{th} quadrotor.
$\Phi_i(t)$	(rad)	The azimuthal angle in spherical coordinate system defining the position of the i^{th} quadrotor.
$\dot{\Phi}_i^{\max}$	(rad/s)	The first derivative of the azimuthal angle of the i^{th} quadrotor.
$\ddot{\Phi}_i^{\max}$	(rad/s ²)	The second derivative of the azimuthal angle of the i^{th} quadrotor.
R_0	-	The rotation matrix demonstrating the attitude of the payload.
ρ_i	-	The vector pointing from the mass center of the payload towards the connection point of the i^{th} quadrotor's cable and the payload.
l_i	(m)	The length of the i^{th} cable.
N_q	-	The number of quadrotors.
$t^{\min}$	(s)	The least time for finishing the mission.
$t^{\max}$	(s)	The most time for finishing the mission.

**FIGURE 3.** An example of the penalty distribution in different slices of the environment using the introduced obstacle modeling.**FIGURE 4.** An example of the risk distribution in the different slices of the environment using the introduced risk sources modeling.

III. FUZZY HYBRID WOA-PSO ALGORITHM

The proposed algorithm in this paper consists of the WOA and PSO algorithms. The WOA algorithm has received lots of attention in recent years. It also has been successfully used for solving many optimization problems. Also, researchers in many scientific and engineering applications have used the PSO algorithm. There are many improved versions of it,

which have enhanced the performance of the conventional PSO algorithm. This paper proposes an improved Fuzzy Hybrid WOA-PSO algorithm, which we will call as Fuzzy-WOAPSO algorithm. The proposed algorithm uses the WOA and PSO algorithms in parallel with some improvements. The WOA [46] and PSO [54] algorithms are among the well-known optimization algorithms, so we do not describe

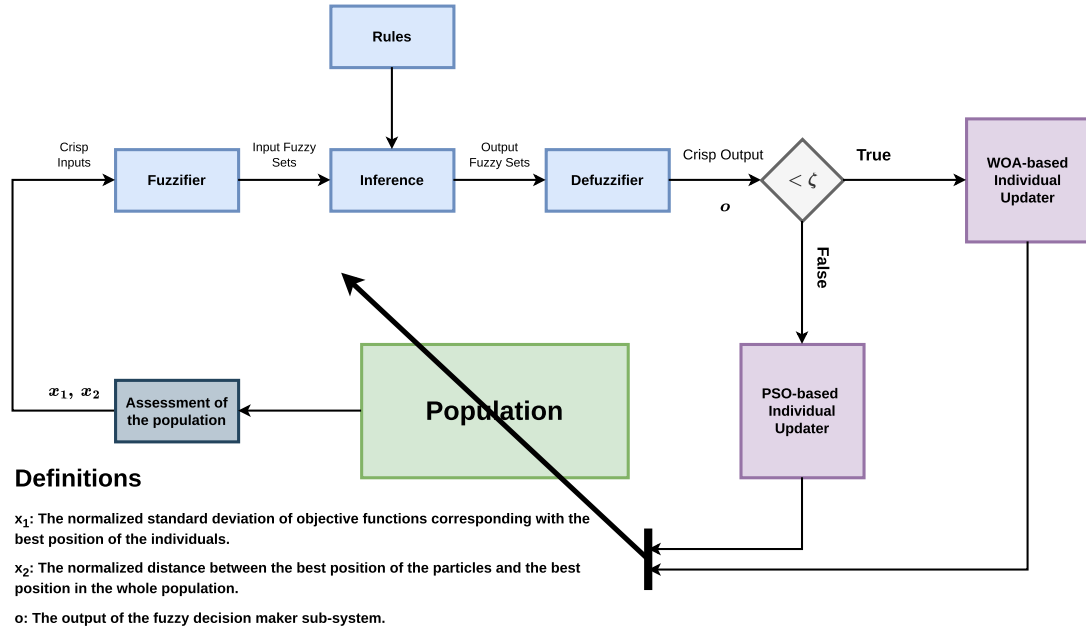


FIGURE 5. Block-diagram representation of the proposed optimization algorithm.

them and their formulation in detail. We have represented the overall process in the proposed algorithm as a block-diagram in Figure 5. In this section, we will discuss how to improve these two algorithms and combine them to work together.

We enhance the WOA in the proposed algorithm by setting the b time variable. b determines the shape of the logarithmic spiral in the bubble-net attacking mechanism. The value of b is decreased uniformly from b_u to b_l over the iterations. These two values should be selected based on the problem under study.

Also, we improve the PSO algorithm by storing the best velocity vector for each particle and the whole population. We use these vectors for updating the velocity vector while evolving the population. The new formula for updating the particles' position is as below:

$$v_i \leftarrow \omega v_i + \phi_p r_p (x_i^b - x_i) + \phi_g r_g (x^g - x_i) + \phi_p^v v_i^b + \phi_g^v v^g \quad (24)$$

$$x_i \leftarrow x_i + v_i \quad (25)$$

In which v_i^b and v^g are the best velocity of the i^{th} particle and the best velocity of the whole population, respectively. Also, ϕ_p^v and ϕ_g^v are corresponding coefficients that can be obtained by trial and error. In the PSO algorithm, particles update their velocities and positions based on their own best-known position and the best-known position of the entire swarm. By storing and utilizing the best velocities, the algorithm effectively enhances both the exploration and exploitation phases. This mechanism allows particles to maintain momentum towards promising regions of the search space, reducing the likelihood of premature convergence. While this approach might cause the swarm to initially pass

over sub-optimal solutions, it ultimately results in more potent exploration. In practice, this contributes to more efficient convergence by guiding the swarm more directly towards the global optimum, as reflected in the quicker convergence rates observed in the reported results. Another improvement concerning the PSO algorithm is introducing a random walk-based mutation operator, which affects the best position of particles in each iteration. The steps of the random walk-based mutation are obtained by the chaotic map achieved by solving the n-body motion problem caused by gravitation force. The force applied by the j^{th} particle to the i^{th} particle can be represented as below:

$$F_{ij} = G m_i m_j \frac{x_j - x_i}{\|x_j - x_i\|} \quad (26)$$

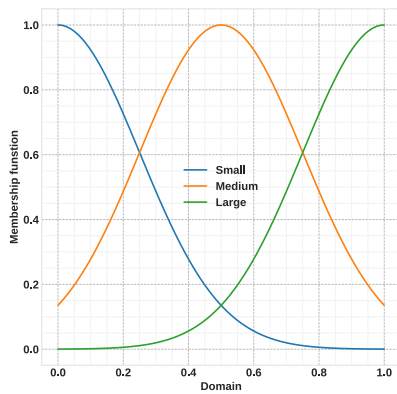
In which m_i is the mass of the i^{th} particle, and G is the gravitational constant.

In each iteration of the proposed algorithm, a fuzzy sub-system decides whether apply WOA or PSO algorithms to a specific individual (particle/whale) in the population. The fuzzy sub-system for this purpose is a Mamdani type 1 fuzzy logic system with a Product inference engine and center of gravity defuzzification method. It has two inputs which are the normalized standard deviation of objective functions corresponding with the best position of the individuals (x_1) and the normalized distance between the best position of the particles and the best position in the whole population (x_2). The output of the fuzzy sub-system, which we will call o , will be compared with a threshold (ζ) and specify the algorithm which will be applied to the individuals (particles/whales). If o is less than ζ , WOA, and if o is more than ζ , PSO will be applied. Table 3 represents the rule base of the fuzzy

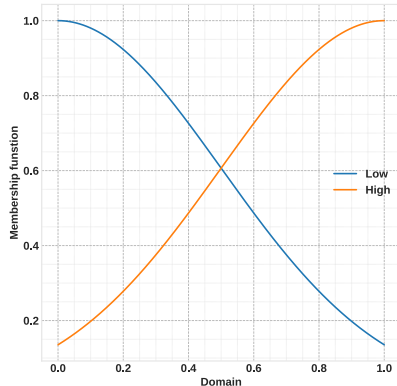
TABLE 3. The rule-base of the fuzzy sub-system.

x_1	x_2	o
Small	Small	Low
Small	Medium	Low
Small	Large	Low
Medium	Small	Low
Medium	Medium	High
Medium	Large	High
Large	Small	Low
Large	Medium	High
Large	Large	High

sub-system. Also, Fig. 6(a) demonstrates the input sets Small, Medium, and Large, and Fig. 6(b) depicts the output sets Low and High. As can be seen, all the sets are of Gaussian type. The input sets have a 0.10 standard deviation, while for the output sets, it is 0.25.



(a) The input sets of the fuzzy sub-system.



(b) The output sets of the fuzzy sub-system.

FIGURE 6. The input and output sets of the fuzzy sub-system.

Notice that while using the improved WOA and PSO algorithms cooperatively, the parameter selection would differ from when the algorithms are used alone. For example, normally ω , and ϕ_p parameters of the PSO algorithm are much less than the ϕ_g parameter. But while using the proposed method, the values of these parameters are selected a little closer to each other, which causes a reduction in the

effect of the best position of the whole population while relocating the particles.

A. SOLVING PATH PLANNING PROBLEM USING THE PROPOSED ALGORITHM

As described previously, the objective function of the optimization is the addition of the path length integral, risk integral, and penalty functions, which can be represented as follows:

$$f = \alpha \int_{t_0}^{t_e} \gamma(\dot{x}_0(t), t) dt + (1 - \alpha) \int_{t_0}^{t_e} \|\dot{x}_0\| dt + \text{Penalty values} \quad (27)$$

In this formula, α indicates the weights of different objectives in the weighted sum method for handling multi-objective optimization problems. Since we have only two objective functions, the weight of one will be α and the other $1 - \alpha$.

The principal strategy for handling the constraints of the defined optimization problem is using the penalty values. In this case, a quadratic function yields the penalty corresponding to each constraint violation.

$$P(v) = c_1 v^2 + c_2 v + c_3 \quad (28)$$

The variable v is always non-negative, as it represents amplitude of violation of the speed, acceleration, altitude, etc. Therefore, the quadratic penalty function $P(v)$, will also always be non-negative. This ensures that the penalty contributes positively or remains zero, in line with the minimization objective.

We use the first type Chebyshev polynomials to model the paths in the proposed algorithm. These polynomials can be represented recursively by the formulas below:

$$\begin{aligned} T_0(x) &= 1 \\ T_1(x) &= x \\ T_{n+1}(x) &= 2xT_n(x) - T_{n-1}(x). \end{aligned} \quad (29)$$

The next important issue before starting the optimization process is defining the solution vector or final decision variables of the problem. Remember that any path in the 3D space consists of three functions indicating the x , y , and z coordinates. Let's assume that we represent every function as a Chebyshev expansion with the first N_c number of polynomials. In the problem, defined in Table 1, the functions that we should determine are:

- x coordinate of the load path
- y coordinate of the load path
- z coordinate of the load path
- ϕ angle of the i^{th} UAV
- θ angle of the i^{th} UAV

Also, we need to determine the end time of the maneuver, t_e . Based on these considerations, the solution vector of the optimization problem would be as below:

$$\begin{bmatrix} c_0^x, c_1^x, \dots, c_{N_c}^x, \\ c_0^y, c_1^y, \dots, c_{N_c}^y, \end{bmatrix}$$

$$\begin{aligned}
& c_0^z, c_1^z, \dots, c_{N_c}^z, \\
& \phi_0^1, \phi_1^1, \dots, \phi_{N_c}^1, \\
& \theta_0^1, \theta_1^1, \dots, \theta_{N_c}^1, \\
& \dots, \\
& \phi_0^{N_q}, \phi_1^{N_q}, \dots, \phi_{N_c}^{N_q}, \\
& \theta_0^{N_q}, \theta_1^{N_q}, \dots, \theta_{N_c}^{N_q}, t_e]^T
\end{aligned} \quad (30)$$

So, the solution vector of the optimization problem should have $(3 + 2N_q)N_c + 1$ elements. Selecting the N_c is a problem-dependent task. If many obstacles and danger zones crowd the workspace and we expect several diversions to be required, N_c must be assumed bigger.

IV. SIMULATIONS

The ultimate objectives of the simulations are to minimize both the path length and the path risk while fulfilling various constraints within the environment. To achieve these goals, the proposed hybrid fuzzy WOA-PSO algorithm optimizes the path planning problem by balancing the trade-off between shorter paths and safer routes through risk-prone areas. Additionally, penalties are applied to the objective function to ensure that all constraints are satisfied, leading to paths that are both efficient and compliant with the specified requirements. The performance of the algorithm is evaluated based on these key metrics: path length, path risk, and the ability to meet all imposed constraints.

In our simulations, we are dealing with a fire incident in a forest. One of the most crucial tasks in such situations is to be ready to deliver emergency supplies to save the lives of firefighters. In the scenarios, the starting point of the cooperative load transportation system is $(-100, -100, 0)$, and the destination of the load transportation is $(100, 100, 25)$. We assume the distribution of the risk caused by the fire time-variant. In real situations, the direction and intensity of the wind, along with factors such as the distribution of flammable materials, move the fire epicenter in different directions. Therefore, this assumption will bring the problem modeling closer to reality. For having a better insight of the problem we study the problem in two conditions, with and without considering the risk. So, the effect of the risk on path planning will be demonstrated based on the achieved results. The first scenario studies the 3D path planning problem considering only path length as the objective. In the second scenario, we add a fire incident in the environment and include the risk as the second objective function.

We have utilized GA [43], PSO [42], GWO [45], WOA [46], FFA [44], HEO [47], IPSO [48], and Fuzzy-WOAPSO to solve the optimization problem. Table 4 represents the parameters of different algorithms. We use a standard notation to name the parameters for all the mentioned algorithms except for GA. Assuming the GA parameters, the $N_{m,GA}$ and $N_{c,GA}$ indicate the number of mutation operators and the number of combinations done in each iteration, respectively. Also, TP_{GA} represents the top

TABLE 4. Parameters used for each of the optimization algorithms.

Algorithm	Parameters
PSO [42]	$\omega = 0.3, \phi_p = 0.3, \phi_g = 2.1$
GA [43]	$N_{m,GA} = 500, N_{c,GA} = 100, TP_{GA} = 10\%$
FFA [44]	$\gamma = 2.5 \times 10^{-4}$
GWO [45]	$a: 2 \rightarrow 0$
WOA [46]	$b = 0.1$
HEO [47]	Leaders: top 40% individuals, Explorers: individuals in top 40% – 80%, Followers: individuals in top 80% – 90%, Losers: the remaining individuals
IPSO [48]	$c_1 \in (2.75, 1.75), c_2 \in (0.5, 2.25)$ $e = 6, h = \text{atan}(20 - e) + \text{atan}(e)$ S-Shaped function adjustment of $\omega \in (0.1, 0.85), o = 13$
FuzzyWOAPSO	$\omega = 0.3, \phi_p = 0.3, \phi_g = 1.5, b_u = 1.0 \times 10^{-1}, b_l = 6.6 \times 10^{-4}, \phi_p^v = 0.3, \phi_g^v = 1.5, \zeta = 0.5$

percent of the population. Notice that we have applied the mutation operator only to elites in the population. We have adopted this enhanced GA from the reference [43]. We consider all the algorithms to consist of 100 individuals, and for each run, we evolve the population in 100 iterations. The parameters for each algorithm have been carefully selected based on extensive testing and experimentation with various values to achieve optimal results. Additionally, the chosen values align with standard parameters commonly reported in the literature, ensuring consistency with best practices. These parameters significantly influence the performance of the algorithms, and their selection has been made with careful consideration to balance exploration and exploitation within the optimization process.

As is said, we handle the constraints of the introduced optimization algorithm using penalty values. These penalties are in integral and quadratic forms. The quadratic penalty forms were chosen due to their simplicity and effectiveness in enforcing constraints within the optimization process. Quadratic penalties ensure that small violations of constraints lead to mild penalties, while larger violations result in more significant penalties, providing a smooth gradient that aids in guiding the optimization toward feasible solutions. The coefficients of the quadratic penalty functions used in the simulation are as follows. In the case of starting and ending point constraints, we define the penalty function as below:

$$P(v) = v^2 + 10v \quad (31)$$

For the altitude limit constraint, we define the penalty function as:

$$P(v) = 100v^2 + 1000v \quad (32)$$

As can be seen, according to the coefficients of the penalty function, the altitude limit in our case study is considered more important than the other constraint. The speed and acceleration penalties are defined as follows:

$$P(v) = v^2 + 100v \quad (33)$$

The $\phi_i(t)$ and $\theta_i(t)$ angles are limited to their defined intervals by using trigonometric functions. Let's assume that $\phi_i^{\text{ch}}(t)$ is a Chebyshev function consisting of a summation of Chebyshev polynomials. For calculating the $\phi_i(t)$ using the $\phi_i^{\text{ch}}(t)$ the formula below can be used:

$$\phi_i(t) = \frac{\pi}{4} \sin(\phi_i^{\text{ch}}(t)) + \frac{\pi}{4} \quad (34)$$

Similarly, based on the $\theta_i^{\text{ch}}(t)$, $\theta_i(t)$ can be calculated as below:

$$\theta_i(t) = \pi \sin(\theta_i^{\text{ch}}(t)) + \pi \quad (35)$$

Also, notice that t_0 is assumed to be zero. So we can write the mission time constraint as below:

$$t^{\min} \leq t_e \leq t^{\max} \quad (36)$$

Since t_e is determined directly, the optimization algorithm can satisfy this constraint by limiting the parameter.

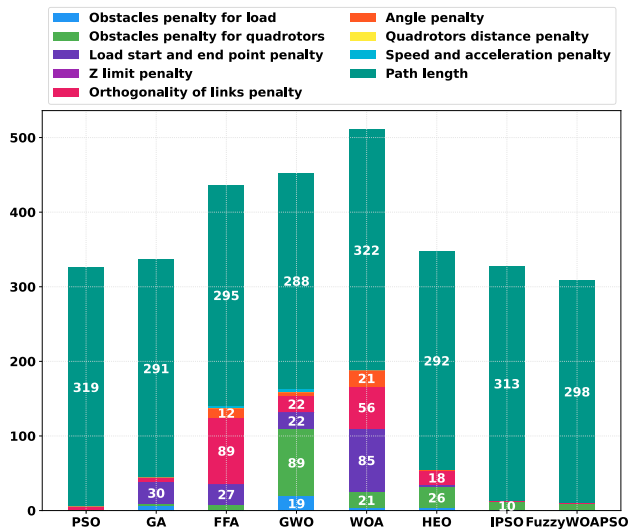


FIGURE 7. Comparison of the components of the objective functions for achieved results by different optimization methods for the first case study.

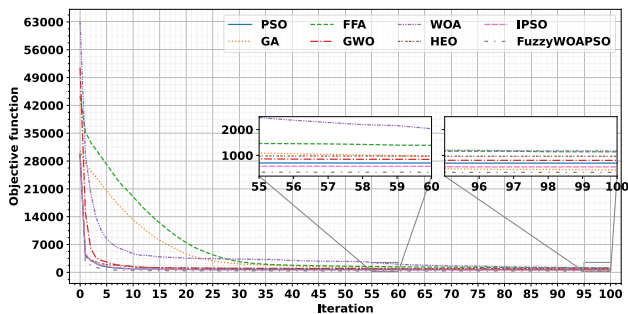


FIGURE 8. Convergence diagram of the optimization algorithms achieved for the first scenario.

The simulations have been performed using C++ codes compiled with G++ 9.3.0 compiler on a Intel(R) Core(TM)

i7-2640M CPU with 8 GB of RAM. Compilation has been done using -O2 optimization parameter, which achieves a relative balance between increasing performance and limiting the impact on compilation time and system resources. The analysis and representation of the outputs are done by Python using the libraries NumPy [55], and Matplotlib [56]. Notice that, the output data in various runs over iterations are available on the GitHub repository dedicated to this paper (Cooperative Load Transportation Path Planning).

A. SCENARIO 1: WITHOUT CONSIDERING RISK

In the first scenario, we assume an environment with obstacles distributed as represented in Figure 3. The only objective of the path planning problem is to achieve the shortest collision-free path in this environment. This case study can evaluate the basic capabilities of the proposed model and different optimization algorithms for solving the conventional path planning problem extended for cooperative cable-suspended load transportation system. The second scenario is more challenging and can reveal the differences between the algorithm better.

The length of the cables used for transporting the load and the diameter of the UAVs are both assumed as 0.5m. Also, the ρ_i s are considered as below:

$$\begin{aligned} \rho_1 &= [+1, +1, 0]^T \\ \rho_2 &= [+1, -1, 0]^T \\ \rho_3 &= [-1, +1, 0]^T \\ \rho_4 &= [-1, -1, 0]^T \end{aligned}$$

We assume all the obstacles as cylindrical with the parameters below:

$$r = 10, h = 200, k = 20, \sigma = 3.$$

The optimization problem introduced in this scenario is the basis for the upcoming one, which is more complicated and challenging. Notice that we limit the arrival time of the load at the destination to the interval [20, 50] seconds in both scenarios. Also, we define other upper and lower limits for parameters as:

$$\begin{aligned} v_0^{\max} &= 10., & \alpha_0^{\max} &= 5. \\ z_0^{\min} &= 0., & z_0^{\max} &= 40. \\ \dot{\phi}_i^{\max} &= 5., & \phi_i^{\max} &= 5. \\ \ddot{\phi}_i^{\max} &= 2.5, & \ddot{\phi}_i^{\max} &= 2.5 \end{aligned}$$

We have obtained the results by running the algorithms 100 times. Table 5 represents the achieved results for the first case study. The results achieved by the proposed algorithm demonstrate a better performance, not only in the shortest path achieved over 100 executions, but also in a statistical sense. The achieved standard deviation of the path length over multiple executions is a factor representing the robustness of the optimization algorithm starting from random initial solutions. As reported in Table 5, the standard deviation value achieved from the results of the proposed algorithm

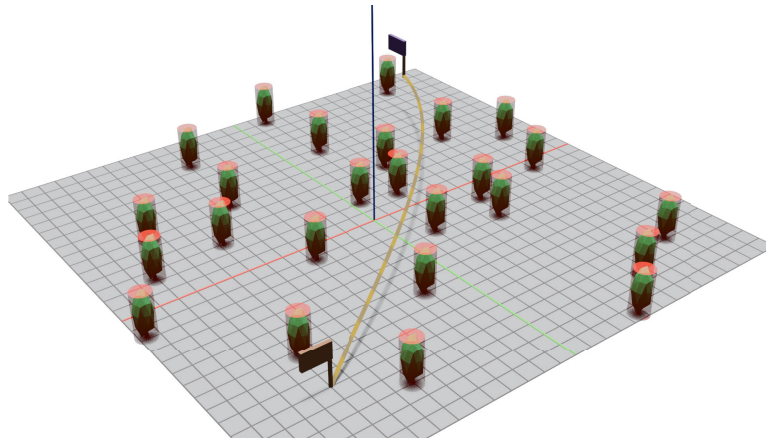


FIGURE 9. 3D illustration of the designed path achieved by the proposed algorithm for the first scenario. The dashed orange curve indicates the optimal path, and the brown and purple flags demonstrate start and finish of the path, respectively.

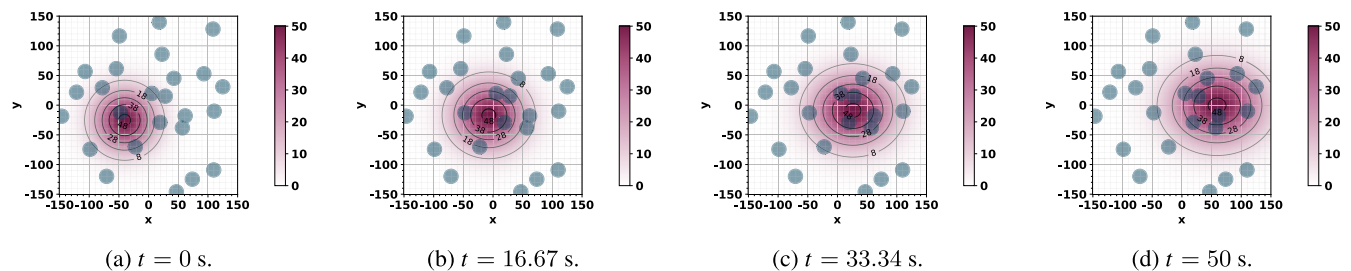


FIGURE 10. Risk distribution in the environment at different times.

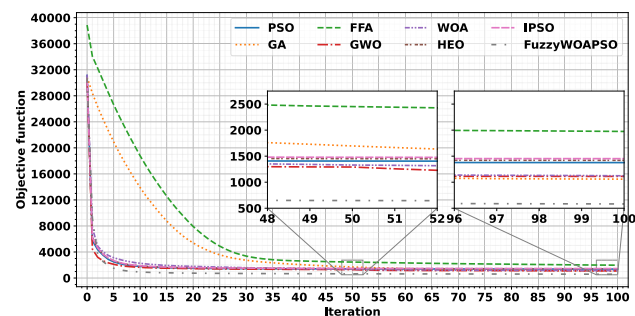


FIGURE 11. Convergence diagram of the optimization algorithms for the second scenario.

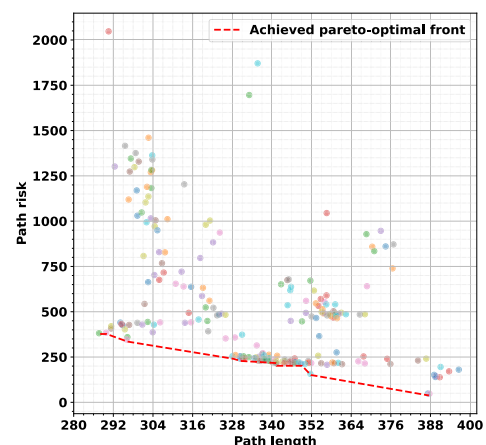


FIGURE 12. Pareto-optimal front achieved by FuzzyWOAPSO algorithm.

over 100 runs is 18.3577, which is significantly less than other algorithms. Such a low standard deviation value along with achieving better minimum path length indicates that the proposed algorithm is capable of providing a better global search and more robust behavior. Also, unlike the results of the other algorithms, the mean value of achieved path lengths by the proposed algorithm (343.8505) is near to the achieved minimum path length (308.8815), which is another indication of the stability of the proposed algorithm considering different random initial solutions. The next stable algorithms, based on the results, are IPSO, GA and then PSO algorithm.

The WOA, which is one of the elements of the proposed algorithm, by achieving a standard deviation of 875.4307 and a mean of 1172.3843, demonstrates a low reliability. So, these results suggest that the hybridization is capable of improving the evolutionary algorithms significantly.

Figure 7 demonstrates the details of the cost function for the best answer obtained from each algorithm. Figure 8 illustrates the convergence diagram of the five algorithms

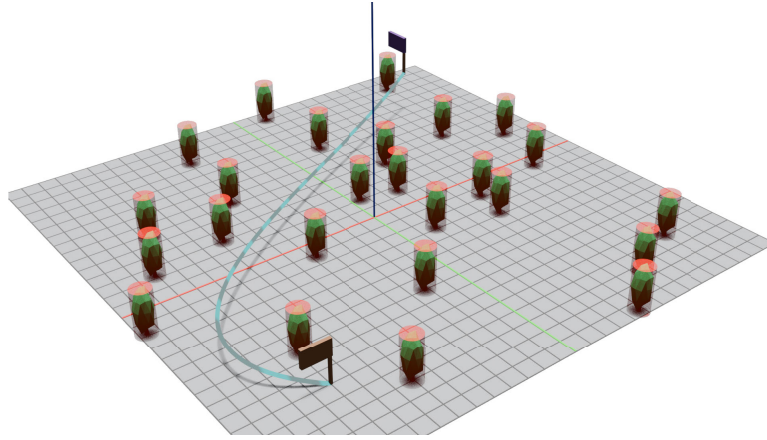


FIGURE 13. 3D illustration of the designed path achieved by the proposed algorithm ($\alpha = 0.5$). The dashed blue curve indicates the optimal path, and the brown and purple flags demonstrate start and finish of the path, respectively.

TABLE 5. Comparison of the results achieved by various algorithms for the first case study.

	PSO	GA	FFA	GWO	WOA	HEO	IPSO	FuzzyWOAPSO
Min	325.4508	336.6487	436.7994	452.4688	510.5924	347.4718	327.37202	308.8815
Mean	701.3949	439.2062	1139.4344	808.1540	1172.3843	967.9839	556.2136	343.8505
Std.	800.2847	159.7858	450.2604	920.1691	875.4307	937.6705	119.0528	18.3577

TABLE 6. Comparison of the average execution time of algorithms.

	PSO	GA	FFA	GWO	WOA	HEO	IPSO	FuzzyWOAPSO
Case 1	48.0401 s	117.6714 s	45.7733 s	38.8305 s	58.4889 s	50.1778 s	52.4762 s	60.2652 s
Case 2	59.8573 s	143.5027 s	60.1121 s	55.3741 s	59.3509 s	59.9968 s	64.3416 s	71.6022 s

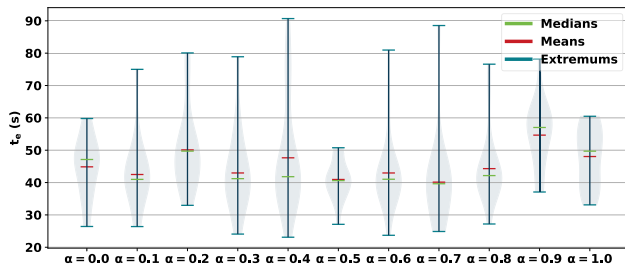


FIGURE 14. A statistical representation of the achieved t_e with different α values over 20 runs.

compared with the proposed algorithm. Notice that we have achieved the convergence diagram by averaging the results of 100 runs for each of the algorithms. Also, Figure 9 represents the 3D illustration of the path and environment. Finally, the elements of the solution vector achieved by the proposed algorithm as an example are as following:

$$S_1 = [1.1098, 107.5807, -1.0994, -7.5582, -12.8679, 103.4527, 12.8681, -3.4402, 16.4441, 11.1356, -3.9492, 1.3476, 5.9907, -5.3364, -6.3525, -3.5493, -0.5747, 5.7878, -0.4671, -1.424,$$

$$\begin{aligned} & -8.7674, 11.151, 2.2924, -1.404, -6.953, \\ & 1.3544, 7.8577, 4.5176, 2.1476, 1.8888, \\ & 2.311, 1.3866, -10.9609, 6.5264, 7.088, \\ & 2.5518, -8.8733, 10.8047, 1.526, -1.3411, \\ & -4.269, 5.0415, 3.2309, 1.1371, 26.7835]^T \end{aligned} \quad (37)$$

Which equation (30) can be used for extracting any parameter in the solution vector.

B. SCENARIO 2: CONSIDERING RISK

In the second scenario, in addition to seeking shortest path, we also consider the path risk as the second objective function. In this scenario, the risk distribution changes over time in the environment, meaning that the fire spreads in the forest and moves to other areas. In other words, the time to cross any place will be very critical. The position and shape of the risk distribution for various time instances after starting the task are depicted in Figure 10, and the parameters of the risk sources are as below:

$$\begin{aligned} x_r(t) &= -40. + 2 \times t, \quad c_x(t) = 50. + 0.5 \times t \\ y_r(t) &= -25. + 0.5 \times t, \quad c_y(t) = 50. + 0.25 \times t \end{aligned}$$

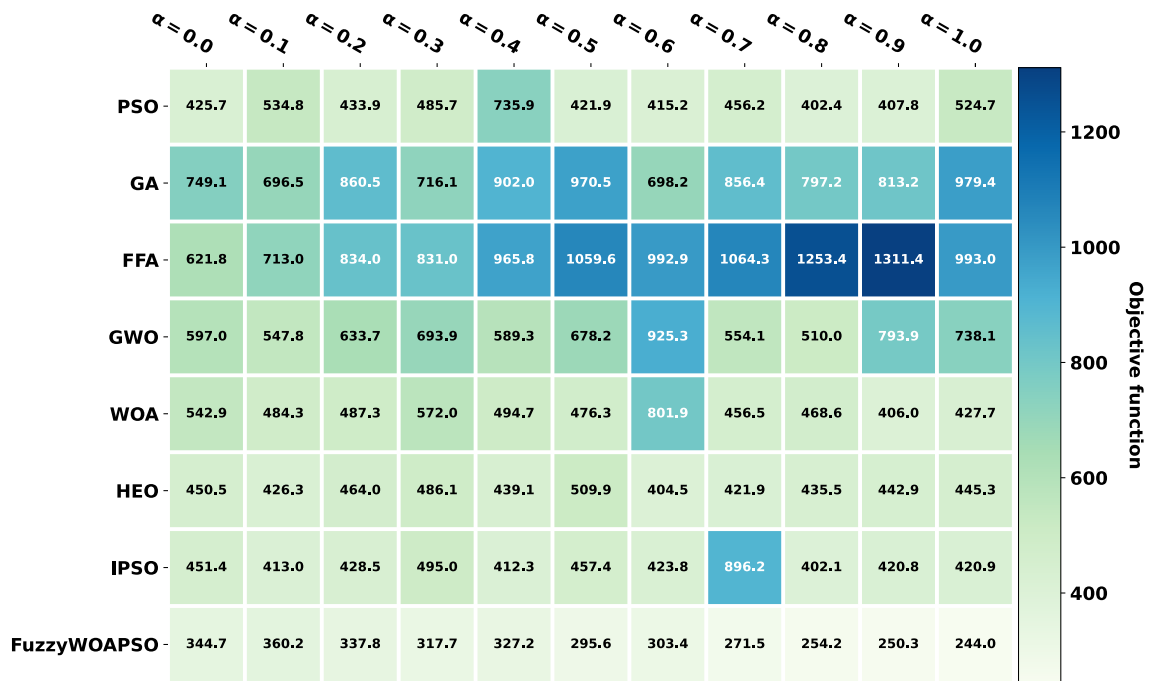


FIGURE 15. Comparison of the best achieved results by different optimization algorithms.

$$\begin{aligned} z_r(t) &= 25., \quad c_z(t) = 100. \\ k(t) &= 50. \end{aligned} \quad (38)$$

As can be seen in the Figure 10, the position of the risk distribution changes over time, and it expands slowly. It is clear that by increasing the time coefficients in formula (38), the risk distribution will move and expand faster.

Figure 11 represents the convergence diagram of the optimization problem, which is the average of 20 runs for $\alpha = 0.0, 0.1, \dots, 1.0$. As can be seen, the proposed algorithm has experienced a better convergence reaching a significantly better solutions compared with other algorithms just over 50 iterations. Figure 12 depicts the Pareto-optimal front obtained by solving the optimization problem using the proposed FuzzyWOAPSO algorithm. Considering the objective functions plane, there are approximately $20 \times 11 = 220$ points. Notice that we have eliminated some solutions that are far from the majority of the other solutions by limiting horizontal and vertical axes, which gives us a better concentrated view of the solutions. Figure 13 depicts the 3D representation of the achieved path for load in the forest when $\alpha = 0.5$. Figure 14 represents the violin diagram of the mission times obtained using the proposed algorithm. Figure 15 represents the results achieved by different optimization algorithms compared with each other for different α values. As can be seen, for all α values, the proposed algorithm has achieved a better solution with considerably better performance. Considering $\alpha = 0$, the generated path by the proposed algorithm has an objective function equal 344.748. The second best performance in this case is achieved

by the PSO algorithm with a objective function of 425.747, which means that the proposed algorithm was able to achieve a result approximately 19.02% less than best other achieved result. This factor for $\alpha = 0.5$ and $\alpha = 1$ is 29.99% and 42.03% compared with PSO and IPSO algorithms as the second best results, respectively. According to these results, the proposed algorithm was able to overcome the other algorithms.

Finally, Table 6 provides the average execution time of different algorithms for all three case studies. As can be seen, the proposed FuzzyWOAPSO algorithm yields considerably better results compared with the other algorithms without additional computational effort. The computational cost of all algorithms are affected by the number of obstacles in the region. Also, each algorithm has its specific computational efficiency which determines the difference in execution time with other algorithms.

V. CONCLUSION

We studied the path planning problem for cooperative cable-suspended load transportation. For this purpose, we have proposed a multi-objective model of the path planning problem and reformulated it as a single-objective optimization problem using the weighted sum method. We have also proposed a novel enhanced fuzzy hybrid optimization method based on the WOA and PSO algorithms, incorporating unique enhancements such as a chaotic walk-based mutation operator and the strategic use of stored velocities, to solve the path planning problem. We used the proposed algorithm to solve a realistic case study and compared the results with

seven other well-known algorithms. The proposed model can represent real-world problems and be used in practical applications. The simulation results prove the abilities of the proposed model. The results confirm that the proposed FuzzyWOAPSO algorithm can effectively solve the path planning problem for a cooperative load transportation system compared with other algorithms. Future work will focus on several key areas to build upon the findings of this study. First, we plan to extend the current path planning model to a robust optimization problem, ensuring greater resilience and performance under varying and uncertain conditions. Second, we aim to address dynamic path planning and real-time adaptation, enabling the system to respond effectively to real-time environmental changes and unforeseen obstacles. Finally, we will study the scalability of the model, evaluating its efficiency and effectiveness when applied to larger fleets of UAVs and more complex logistics networks. These advancements will enhance the robustness, adaptability, and scalability of cooperative load transportation systems, paving the way for broader practical applications. As a future work, we would address online path-planning and control problems and integrate them with the results achieved by offline path-planning method proposed in this paper to have a complete framework of cooperative cable-suspended path-planning ready for being used in practical applications.

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AMIR ARSLAN HAGHRAH was born in Tabriz, Iran, in 1992. He received the B.Sc. degree in control engineering from the University of Tabriz and the M.Sc. degree in control engineering from the K. N. Toosi University of Technology, Tehran. He is currently pursuing the Ph.D. degree in control engineering with the University of Tabriz. His research interests include soft computing and intelligent systems, cooperative robotics, and linguistic characterization of natural data.



SEHRANEH GHAEMI received the B.Sc. degree in power engineering from Amir Kabir University, Tehran, Iran, in 1999, and the master's and Ph.D. degrees in control engineering from the University of Tabriz, Tabriz, Iran, in 2003 and 2010, respectively. She is currently a Professor with the Control Department, Faculty of Electrical and Computer Engineering, University of Tabriz. Her main research interests include fuzzy systems, evolutionary algorithms, and intelligent control.



MOHAMMAD ALI BADAMCHIZADEH was born in Tabriz, Iran, in 1975. He received the B.S. degree in electrical engineering and the M.S. and Ph.D. degrees in control engineering from the University of Tabriz, Tabriz, in 1998, 2001, and 2007, respectively. He is currently a Professor with the Faculty of Electrical and Computer Engineering, University of Tabriz. His current research interests include adaptive control, stability of systems, and system identification.

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